



EE200: Audio Fingerprinting

SIGNALS, SYSTEMS & NETWORKS • SHAZAM-STYLE IDENTIFIER

Index a library of songs as spectrogram fingerprints, then identify any short clip against it.

◆ LIBRARY ⦿ IDENTIFY ☰ BATCH

SEARCH

Identify a clip

📁 Upload

Drop / pick a file • WAV, MP3, FLAC, OGG, M4A

Record

OR TRY A SAMPLE

sample1

Try

sample2

Try

sample3

Try

sample4

Try

sample5

Try

Identify

①

SPECTROGRAM

②

CONSTELLATION

③

HASHING

④

DB LOOKUP

⑤

SCORING

194 ms

513×386

34 ms

341 peaks

200 ms

4,988 hashes

1988 ms

50 tracks

0 ms

offset 1184

total 2416 ms

MATCH FOUND

Back In The U.S.S.R.

cluster score **1,798** · **163.5×** the runner-up

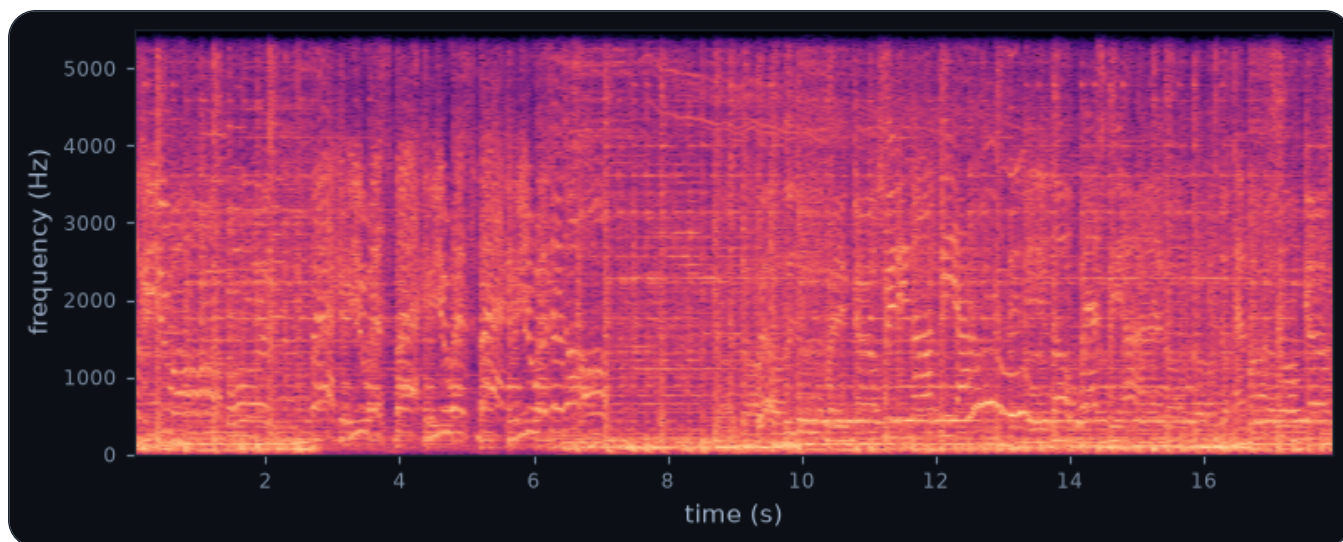
CANDIDATE SCORES

Back In The U.S.S.R.	1,798
Never Gonna Give You Up	11
Under Pressure	7
I Saw Her Standing There	7
Get Back	6
Love Me Do	6

STEP 1 · THE PICTURE

Spectrogram of the clip

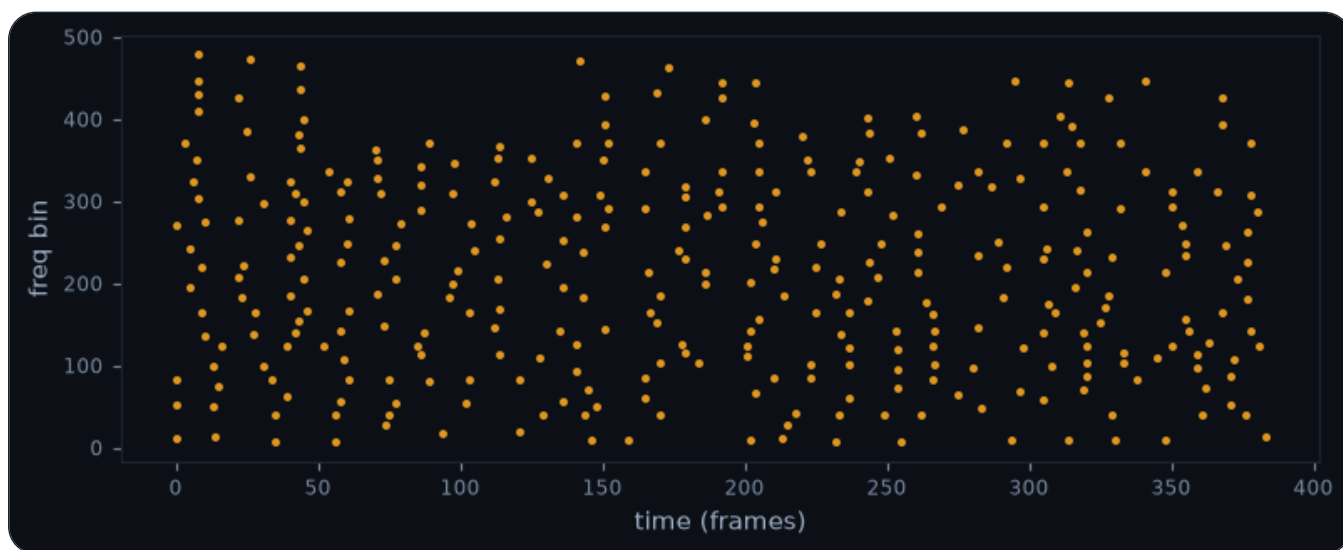
A short sliding window turns the waveform into a time–frequency image — which frequencies sound, and when.



STEP 2 • THE FINGERPRINT

Constellation of peaks

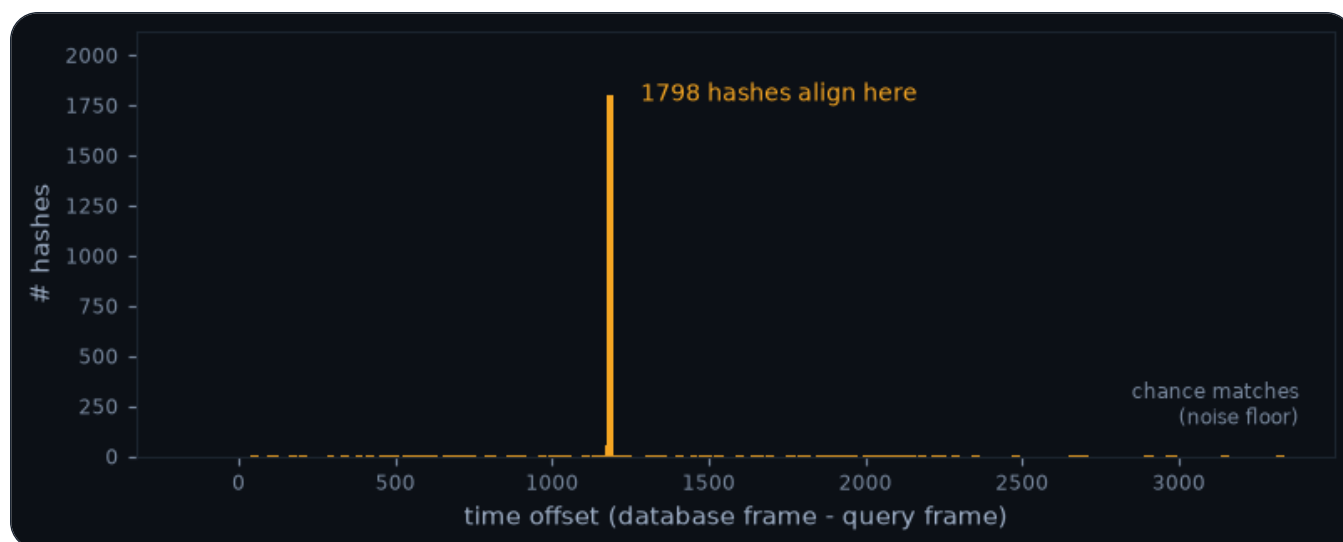
Keep only the loudest local maxima. These sparse points survive noise and become the clip's fingerprint.



STEP 3 • THE PROOF

The alignment spike

Every matched hash votes for a time offset (database frame – query frame). Chance matches scatter into a flat noise floor; a genuine match makes them **converge on a single offset**. That spike cannot be a coincidence.



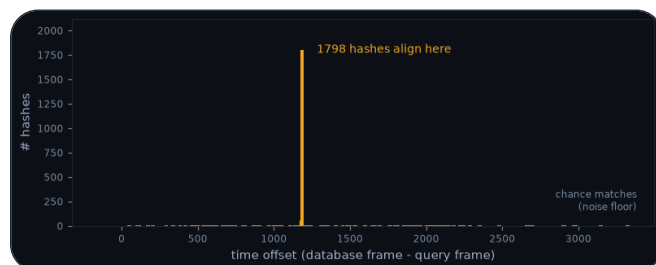
DEEPER • WHY PAIR PEAKS?

Single peaks vs. paired hashes

A lone frequency peak recurs across countless songs, so single-peak matching is noisy. Pairing two peaks with their time gap into $(f_1, f_2, \Delta t)$ is far rarer to collide on — the alignment spike sharpens dramatically. Compare them on this very clip:

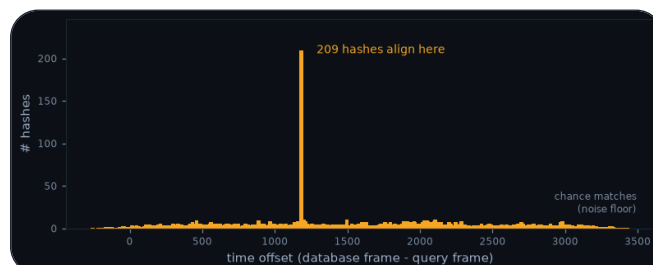
PAIRED HASHES

Back In The U.S.S.R. — score 1,798, confidence 163.5×



SINGLE PEAKS

Back In The U.S.S.R. — score 209, confidence 20.9×



DEEPER • ROBUSTNESS LAB

How far can we push it?

Re-test the same clip under increasing white noise and pitch shift. Watch the score survive heavy noise (peaks are sparse and loud) but collapse under even a small pitch shift (hashes key on *absolute* frequency bins). Green = still correct, red = match lost.

Stress-test this clip

NOISE SWEEP



PITCH SWEEP



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